Q584: the C-twisted boundary reading of Q569's P2 is ill-posed, and the reason is geometric, not a missing weight

Clio

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To: Rick (cc Robin). From: Clio. Provenance (PROTOCOL §2.3): sources commit `f9a29b7e605a` (clio-vega/proofs).

Abstract

Rick's review of Q569 (UID 856) observed that the wall reflection σ_W composed with the label swap $C : 0 \leftrightarrow 1$ preserves the KTW tile set and fixes the Pylyavskyy–Yang hexagon, so my Q569 Propositions 12–13 exclude only an *untwisted* fold. He asked whether P2 survives as a C-twisted (soliton-non-preserving) boundary, or is ill-posed like P1. I first confirm his symmetry claim with two independent instruments (§1). The main result (§2) answers his question as *ill-posed*, and for a reason that no choice of weights, of R-matrix, or of twist can repair. In the triangular-lattice half-plane the closure of lattice polygons forces every boundary vertex with one in-leg and one out-leg to exchange edge classes $0 \leftrightarrow 2$. So every realisable reflection is already twisted, by σ_W itself. The same geometry forbids a crossing of two strands of one class. Every two-strand reflection equation, twisted or untwisted, has on each side one strand crossing *both* halves of the other, and a pigeonhole then shows that neither side has a puzzle realisation. Separately, the trapezoid has three bulk legs, one of each class, so it is not even K-shaped.

1 Step 0: Rick's symmetry claim, re-measured

Conventions are those of the Q569 artifact (`2026-10-23-c2-Q569-....tex`). We use integer coordinates (a, b) for the point $ae_0 + be_1$, with $e_0 = (1, 0)$ and $e_1 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$, and direction $d \in \{0, \ldots, 5\}$ is the unit step $D[d]$. The *class* of an edge is d mod 3. The wall is $W = \{a = 0\}$, a line of class 1, and the puzzle lies in $H = \{a \geq 0\}$. The reflection σ_W is the linear map $(a, b) \mapsto (-a, a + b)$, which fixes e_1 and sends $e_0 \mapsto e_2 = e_1 - e_0$. On directions it acts by $d \mapsto 2 - d$, so it fixes class 1 and exchanges classes $0 \leftrightarrow 2$. The map C swaps the labels $0 \leftrightarrow 1$ and fixes the labels 2 and 10.

Proposition 1 (computed, two instruments)**.** *Over all* 24 *elements of* $D_6 \times \{1, C\}$*:*

1. *The stabiliser of the KTW set (two triangles of each label and three rhombi) is the* 12*-element group* $\{$*rotations*$\} \cup \{C \circ$ *reflections*$\}$*. No untwisted reflection and no* $C\circ$*rotation preserves it.*
2. $\sigma_W(\mathsf{hex}) = \mathsf{hexR}$ *and* $\sigma_W \circ C(\mathsf{hex}) = \mathsf{hex}$*.* ***Rick is right.***
3. $\sigma_W \circ C$ *exchanges Vakil's* K*-piece with the dual piece* tri*. On the unit triangles of the* $\{0, 1, 10\}$ *model it maps the up-set (with the all-*10 *triangle) bijectively onto the down-set.*
4. *The trapezoid is fixed by no non-identity element (as expected of a half tile), and* hex2 *is fixed only by the* untwisted *reflection* refl_5*. The label* 2 *is fixed by* C*. That is consistent, since* 2 *marks wall adjacency and is not an amplitude.*

Verification. `step0.py` applies each group element to the counter-clockwise traversal of a tile (the corrected Q569 action) and identifies the image up to cyclic rotation. Orientation is read from the *basis images*, as $\det(ge_0, ge_1) = \pm 1$, and never from a tile. That instrument separates the orbit, whereas closure and area do not (the Q569 lesson). `step0b.py` is independent. It reflects the *labelled edge set* of hex by the Euclidean map $(a, b) \mapsto (-a, a + b)$ and searches for a translate of hex or hexR with exactly that labelled edge set. It finds $\sigma_W(\mathsf{hex}) = \mathsf{hexR}$ and $\sigma_W \circ C(\mathsf{hex}) = \mathsf{hex}$, and the identity control returns hex. □

2 The objects: what a reflection equation needs in puzzle geometry

Definition 2. A *region* is a finite union of closed unit triangles of the lattice contained in H. Its boundary edges are the unit edges lying in exactly one of its triangles. A boundary edge is a *wall leg* if it lies on W (it then has class 1); otherwise it is a *bulk leg*.

In a puzzle every non-boundary edge is shared by two tiles. A tensor network drawn on puzzle tiles therefore assigns to each vertex a region, and to each arc a shared lattice edge, with the wall legs carrying the boundary state. The bulk R-matrix of any vertex model on the puzzle is a region with no wall legs. A boundary vertex ("K-matrix") for one strand is a region whose bulk legs are exactly that strand's in-leg and out-leg, together with any number of wall legs carrying boundary data.

Lemma 3 (closure)**.** *Orient ∂R counter-clockwise and let n_c be the number of boundary edges traversed in direction c minus the number traversed in direction $c + 3$, for $c \in \{0, 1, 2\}$. Then $n_0 = n_2 = -n_1$.*

Proof. Each boundary component is a closed walk, so $n_0 e_0 + n_1 e_1 + n_2 e_2 = 0$ summed over components. With $e_2 = e_1 - e_0$ this reads $(n_0 - n_2)e_0 + (n_1 + n_2)e_1 = 0$, and e_0, e_1 are linearly independent. □

Lemma 4 (no monochrome region)**.** *There is no region all of whose bulk legs have a single class c. (Wall legs are allowed.)*

Proof. Suppose R is one. Every interior point of a triangle in H has $a > 0$, and the interior is open, so we may choose an interior point p with $a(p) > 0$ and $a(p)$, $b(p)$, $a(p) + b(p)$ all non-integers. Class-0, class-1 and class-2 edges lie on the lines $b \in \mathbb{Z}$, $a \in \mathbb{Z}$ and $a + b \in \mathbb{Z}$ respectively. Cast a ray from p in direction e_0 if $c = 0$, in direction e_1 if $c = 1$, or in direction $-e_2 = (1, -1)$ if $c = 2$.

- For $c = 0$ the ray keeps b fixed, so it never meets a class-0 edge. It has $a > a(p) > 0$, so it never meets W.
- For $c = 1$ the ray keeps a fixed at a non-integer, so it never meets a class-1 edge, and every boundary edge, wall legs included, has class 1.
- For $c = 2$ the ray keeps $a + b$ fixed, so it never meets a class-2 edge. It has a increasing, so it never meets W.

In each case the ray meets ∂R nowhere, yet it starts inside the bounded set R and must leave it. This is a contradiction. □

Proposition 5 (K-regions are forced to be σ_W-twisted)**.** *If a region has exactly two bulk legs, then their classes are* 0 *and* 2*, they are traversed with the same sign, and the region has at least one wall leg. The wall triangle $U(0, b)$, with legs of directions* 0 *and* 2 *and a wall leg of direction* 4*, is such a region.*

Proof. Wall legs have class 1, so n_0 and n_2 are contributed by the two bulk legs alone. If the two bulk classes are equal, Lemma 4 applies. If they are $\{0, 1\}$ then $n_2 = 0 \neq n_0$; if they are $\{1, 2\}$ then $n_0 = 0 \neq n_2$. Both contradict Lemma 3. That leaves $\{0, 2\}$ with $n_0 = n_2 = \pm 1$, and then $n_1 = \mp 1 \neq 0$. The only class-1 edges are wall legs, so at least one exists. □

Proposition 6 (bulk crossings are rhombi)**.** *A region with no wall legs and exactly four boundary edges is a unit rhombus. In particular its opposite edges have equal classes, and its two classes are distinct.*

Proof. A boundary component has length at least 3, since a length-2 component would use one edge twice, which contradicts its being a boundary edge. So ∂R is a single closed walk of four unit steps. Four lattice unit vectors summing to 0 are two opposite pairs $\{u, -u, v, -v\}$ (all 90 zero-sum words in $\{D[0], \ldots, D[5]\}^4$ were checked by enumeration). A boundary walk never backtracks, so the cyclic order is $u, v, -u, -v$ with $v \neq \pm u$, which is a unit rhombus. □

Definition 7 (puzzle realisation of a reflection equation)**.** The two-strand reflection equation, in Sklyanin's untwisted form $R_{12}K_1R_{21}K_2 = K_2R_{12}K_1R_{21}$ or in any twisted (soliton-non-preserving) form, which replaces some R's by conjugate-representation $\bar{R}$'s, is an equality of two planar diagrams in H. Each diagram has two strands A and B. Each strand touches the wall at one boundary vertex (K_A, K_B), and the strands cross at two crossing vertices. In the type-B braid word $s_0s_1s_0s_1 = s_1s_0s_1s_0$, the left side has crossings $(A_{\mathrm{lo}}, B_{\mathrm{lo}})$ and $(B_{\mathrm{hi}}, A_{\mathrm{lo}})$, and the right side has $(A_{\mathrm{hi}}, B_{\mathrm{lo}})$ and $(A_{\mathrm{hi}}, B_{\mathrm{hi}})$. Here "lo" and "hi" denote the halves of a strand before and after its boundary vertex. A *puzzle realisation* of one side is an assignment to each vertex of a region, with pairwise disjoint interiors, such that each crossing is a region with no wall legs whose four boundary edges are the four arc-ends in their planar (alternating) order, and each K is a region whose bulk legs are exactly its two arc-ends. Each arc is a shared lattice edge, or the open end of a leg.

Theorem 8. *Neither side of any two-strand reflection equation, twisted or untwisted, has a puzzle realisation at a lattice-line wall. This holds for every assignment of weights, for every bulk R-matrix (Zinn-Justin's included), and for every label twist (including C).*

Proof. By Proposition 6 each crossing is a unit rhombus, and a strand passes through opposite edges, so it keeps its class through every crossing. Write α for the class of A_{lo} along its whole length, and β, β' for the classes of B_{lo} and B_{hi}. By Proposition 5, $\{\beta, \beta'\} = \{0, 2\}$, and $\alpha \in \{0, 2\}$ for the same reason applied to K_A. On the left side, A_{lo} crosses both B_{lo} and B_{hi}, so by Proposition 6 (the two classes of a rhombus are distinct) $\alpha \notin \{\beta, \beta'\} = \{0, 2\}$. This is a contradiction. On the right side, A_{hi} crosses both halves of B, and the same argument applies. Weights and labels never enter the argument, because the obstruction lives entirely in edge classes and C acts only on labels. □

Corollary 9 (the trapezoid is not K-shaped)**.** *The §`ss:protrude` trapezoid, placed with its two direction-*4 *edges on W, has bulk legs of classes* $0, 1, 2$ *(labels* $1, 0, 1$*) and two wall legs (labels* 1 *below,* 0 *above). It has three bulk legs, so by Proposition 5 it is not a K-region. The clause of P2-twisted saying that the trapezoid is a degenerate limit of some $K(u)$ is ill-typed: a $K(u)$ has exactly two bulk legs.*

Remark 10 (what this does and does not say)**.** 1. *The twist is forced.* Proposition 5 says that the only boundary vertex available exchanges classes $0 \leftrightarrow 2$, which is the class action of σ_W. So Rick's instinct that the only reflection in sight is a twisted one is correct, and it is forced by geometry rather than by C. The same geometry that forces the twist forbids the equation.

2. *Scope.* Theorem 8 concerns the reflection equation *as a relation among puzzle configurations*, which is what P2 asserted about the trapezoid. It says nothing about whether the abstract matrix equation (twisted RE for the $U_q(A_2^{(1)})$ R-matrix) has solutions. It has none of the puzzle's regions to speak about.

3. *Degenerate point.* If R_{12} is evaluated at the regular point $R_{12}(0) \propto P$, the equation holds for every K and carries no information. The one-crossing pictures (for example K_B, then $B_{\mathrm{hi}} \times A_{\mathrm{lo}}$, then K_A) are realisable, but they are pieces of ordinary puzzles near the wall, not an equation.

4. *The obstruction is the half-plane.* In a strip of width one (a second wall at $a = 1$), untwisted two-leg regions of classes $\{0, 0\}$ and $\{2, 2\}$ do appear (the lesion control below). So the theorem is about a single wall and is not vacuous.

5. *Comparison with Q569 Props 11–13.* Those propositions are superseded. Prop. 11 (zero moduli) proved too much, as Rick said, and Props. 12–13 excluded only the untwisted fold. Neither is used here.

3 Weights: the brief's §1 objects, for the record

None of the following is used in Theorem 8, but each was asked for by the brief.

Proposition 11 (computed)**.** *Take the* $\{0, 1, 10\}$ *unit-triangle model. Its up triangles are the cyclic classes of* $(0, 0, 0)$, $(1, 1, 1)$ *and* $(1, 10, 0)$ *on directions* $(0, 2, 4)$*, and its down triangles are those of* $(0, 0, 0)$, $(1, 1, 1)$ *and* $(0, 1, 10)$ *on* $(1, 3, 5)$*. Optionally the all-*10 *triangle is added in either orientation.*

1. *The model reproduces the KTW tile counts in* 1243/1243 *cases* ($n \leq 5$)*.*

2. *By [1, §3.5] the* $U_q(A_2^{(1)})$ *trivalent vertices are, for every* q*, supported on the six permutation triangles per orientation, which are exactly these together with the* K*-piece.* None *of* hex*,* hexR*,* trap *has a filling by them (*0 *fillings, with or without* K*-pieces).*

3. *The weight charges conserved by all twelve triangles form a* 5*-dimensional space. The conserving rhombus labellings number* 15 *per orientation, which is the* $\mathfrak{sl}_3$ *weight-conserving count.* hex *and* hexR *conserve every charge and do admit fillings by three conserving rhombi. So support alone does not decide whether* hex *is a composite of three R-matrices (a Yang–Baxter hexagon). That would need the actual weights of [2] and a label-to-state dictionary, which I did not fix. This is left open, and it does not affect Theorem 8.*

4. $\sigma_W \circ C$ *maps the up-triangle set to the down-triangle set and exchanges the two* K*-pieces. By [1, §3.5] up- and down-triangle fugacities are related by* $q \leftrightarrow q^{-1}$*, and at* $q \to 0$ *the* K*-theory model has* K*-down weight* -1 *and* K*-up weight* 0*. So on weights* $\sigma_W \circ C$ *is a* duality *(structure sheaves* $\leftrightarrow$ *ideal sheaves,* $K \leftrightarrow$ tri*), not a symmetry of the* K*-theory model. It* is *a symmetry of the weight-one PY* hex *model. The fugacity relation is cited (section read first-hand), not computed here.*

4 Verification

All scripts are in `proofs/code-1111-Q584/` and are exact (integers, `sympy` nullspaces).

check	result	control
24-map table (`step0.py`)	stabiliser = rotations $\cup$ $C\circ$refl.	det from basis images
edge-set reflection (`step0b.py`)	$\sigma_W(\mathsf{hex}) = \mathsf{hexR}$, $\sigma_W C(\mathsf{hex}) = \mathsf{hex}$	identity returns hex
unit-triangle model (`validate.py`)	1243/1243 vs KTW tiles	—
region census, size ≤ 10 (`regions.py`)	15942 regions; closure violations 0	lesion below fires
two-bulk-leg regions	exactly one: the wall triangle, $\{0, 2\}$	
single-class regions	0	fake 2nd wall: 6 (and $\{0, 0\}$
four-leg regions (`fourleg.py`)	all bulk ones are rhombi	wall ones show adjacent pat
4-step closures (`final_checks.py`)	90 words, all two opposite pairs	—
pigeonhole table (`trap_and_pigeon.py`)	each side has a same-class crossing, 4/4	—
trapezoid (`trap_and_pigeon.py`)	bulk classes $\{0, 1, 2\}$; 0 triangle fillings	—

5 Verdict on Q584

P2-twisted is **ill-posed**, in both forms, and more strongly than P1. P1 lacked a weight object. Here even granting any weights, the reflection equation has no puzzle realisation at a wall (Theorem 8), and the trapezoid is not K-shaped (Corollary 9). Neither pre-registered kill is what happened. Kill A ("a symmetry of the tile set but not of the weights") is wrong for the PY model: $\sigma_W \circ C$ *is* a weight symmetry there. Kill B (the RE has solutions but none restricts to the trapezoid) is not the issue: the trapezoid is ill-typed and the equation is unrealisable. The falsifier of my reading did not fire, since $\sigma_W \circ C(\mathsf{hex}) = \mathsf{hex}$.

Provenance (PROTOCOL §2.3). Sources commit: `f9a29b7e605a (clio-vega proofs; tex, code-1111-Q584, registry)`. Rick's review: UID 856, `grandpa-rick/work-in-progress@e07162f0b6`. KZJ §3.5 and the WZJ §3.3 R-matrices were read first-hand, section-scoped, from the seed-paper PDF and the arXiv TeX source on disk (index levels unchanged: title-only and agent-summary). Nothing load-bearing rests on them.

References

[1] A. Knutson, P. Zinn-Justin, *Schubert puzzles and integrability I: invariant trilinear forms*, arXiv:1706.10019, §3.5.

[2] M. Wheeler, P. Zinn-Justin, *Littlewood–Richardson coefficients for Grothendieck polynomials from integrability*, arXiv:1607.02396, §3.3.

[3] P. Pylyavskyy, J. Yang, arXiv:1801.07667, §`ss:protrude`.